

COMP.2560: System Programming Assignment #4: Signals

ASSIGNMENT 4: Overview

The purpose of this assignment is to evaluate your comprehension on Signals.

There are four (4) questions in this assignment, described below.

ASSIGNMENT 4 – Question 0

Please submit the Bash Scripting DataCamp course completion certificate that you should have obtained during the reading week.

ASSIGNMENT 4 – Question 1

Write a C program where the parent process generates exactly two child processes such that:

- the parent process uses the `waitpid(...)` function to wait for the second child process and prints its `pid` and its exit status value (of your choice), and
- the first child process becomes an orphan.

The output of your program should be similar to the one shown below:

```
1st child created.PID=1650893
becoming orphan
2st child created.PID= 1650894
I will be waited for by my parent...
Parent waited for pid=1650894 sucessfully
danwu@delta:~/comp2560w2024/examcode$ 1st child, PID=1650893, PPID=1 orphaned!
```

ASSIGNMENT 4 – Question 2

This question is related to a slide we discussed in class, which is shown below.



Figure 1: `signal()` and `exec()`

Verify its claim by designing a program to demonstrate that:

- In the parent process, you install a handler for `(Ctrl+c)` and you ignore `(Ctrl+z)`.
- The parent process then forks a child process.
- Inside the child process, using `execXX()` function to run a simple program (such as `"donothing.c"` we discussed in class), and check to see if the handler you installed (for `Ctrl+c`) in the parent is set back to the default handler, and
- Check to see if `Ctrl+z` is still ignored

Hint: You may use the `kill()` function to send signals to process to test the claims.

Explain briefly how your program is designed to test the claims **as comments** in your source code.

Script the output from running your program with the timing option.

ASSIGNMENT 4 – Question 3

Write a program simulating an alarm clock:

- In the main function:
 - You fork a child process, and
 - The child process sleeps for 5 seconds (the number of seconds is a command line argument, see below for a sample run), after which
- The child process sends the signal (`SIGALRM`) to its parent process.
- The parent process:
 - After forking the child process, pauses upon receiving the `SIGALRM` signal, and
 - Prints out a message `"Ding!"`.

The following is a sample run:

```
$/alarm 5
Alarm application running waiting for alarm to go off
<5 second pause> Ding!
Done!
```

ASSIGNMENT 4 – Question 4

This exercise shows the idea of how to use one user-defined signal handler to handle two signals. The two signals chosen for this exercise are `SIGINT` and `SIGQUIT`.

Read (or Google) the descriptions of these two signals (pages 319 and 320 in the textbook respectively). Note that this idea applies to other signals as well.

What you need to do is to:

- Write your own one signal handler function, and

- Install this handler for both SIGINT and SIGQUIT.
- When your signal handler is invoked, print out the signal no (which is the argument to your user-defined signal handler function), so that we know which signal has been caught.

ASSIGNMENT 4 – Wrap-up and Submission

- For all questions, you need to submit your source code.
Also, use the script command with the timing option to record you compiling and running your program and submit the script file and timing file for each question as well.
Submit the source code for each question and include any accompanying files (if applicable) needed for that question.
Please make sure all your code works properly on CS Linux server.
 - Submit the link to the video you will create.
 - Please properly name all your files so that markers can easily identify which file is for which question.
See the *Assignment Submission Specification document* for naming files in your submissions for this semester.
 - Record one video with audio showing your source code, you compiling and running each program and explaining how each of the underlined items in each question's description is realized in your code.
For the convenience of marking your video submission, please add chapters to your video so that markers can easily jump to the start of each question in your video. If you do not know how to add chapters to your video, please watch this short video to learn how:
 - <https://www.youtube.com/watch?v=d5uSWaCPLj4>
- If you are using Microsoft Streams, this link shows how to add chapters:
- <https://support.microsoft.com/en-us/office/using-manual-chapters-with-videos-on-stream-on-sharepoint-8bbf61eb-c00b-42b5-a514-cce2e45eb6ea>
- Please note the late assignment submission policy stated in the course syllabus.



ASSIGNMENT 4 – Appendix: Grading Rubric

Q.0		5 Marks
Q.1	- Creation of exactly two child processes: 2 Marks - The parent process: - Uses the waitpid(. .) function to wait for the second child process (2 Marks), and - Prints its pid (1 Mark), and - Extracts its exit status value (2 Marks) Total: (5 Marks) - The first child process becomes an orphan: - Use sleep() or any other mechanism to make sure that the parent process terminates first (2 Marks) - Print out info. (1 Mark) Total: 10 Marks	
Q.2	- Install a handler for (Ctrl+c). (2 marks) - Ignore Ctrl+z (2 Marks) - The parent process forks a child process. (1 Mark) - Use execXX() function to run a simple program (2 marks) - Check to see if the handler you installed (for Ctrl+c) in the parent is set back to the default handler (1 Mark), - Check to see if Ctrl+z is still ignored (1 Mark) - Explain your program using comments in the source code - Script the output from running your program with the timing option (1 Mark) Total: 10 Marks	
Q.3	fork() a child process (1 Mark) - The child process sleeps for 5 seconds (1 mark) - The number of seconds is a command line argument (1 Mark) - The child process sends the signal (SIGALRM) to its parent process (2 Marks) - The parent process, after forking the child process, pauses upon receiving the SIGALRM signal (1 Mark) - The process prints out a message "Ding!" (2 Marks) - Proper design of this signal handler to print Ding! (2 Marks) Total: 10 Marks	
Q.4	- Write your own one signal handler function (2 Marks) - Install this handler for SIGINT (2 Marks) - Install this handler for SIGQUIT (2 Marks) - Invoke the signal handler, print out the signal no for SIGINT (1 Mark) - Invoke the signal handler, print out the signal no for SIGQUIT (1 Mark) Total: 8 Marks	
	Submit video 2 Marks	
		TOTAL: 45 Marks